

Assignment

Data Structure Lab (MCAN-291)

1. Write a program in python to implement following operations on a single linked list.

- a) Insert a node at the end of the linked list.
- b) Insert a node at the front of the linked list.
- c) Delete an element from the linked list.
- d) Display the elements of the linked list.
- e) Count number of elements present in the linked list.
- f) Search whether a given element present in the linked list or not.

2. Write a program in python to implement following operations on a doubly linked list.

- a) Insert a node at the end of the linked list.
- b) Insert a node at the front of the linked list.
- c) Delete an element from the linked list.
- d) Display the elements of the linked list.
- e) Count number of elements present in the linked list.
- f) Search whether a given element present in the linked list or not.

3. Write a program in python to implement following operations on a Stack using array.

- a) Push an element.
- b) Pop an element.
- c) Display the stack.

4. Write a program in python to implement following operations on a Queue using array.

- a) Insert an element.
- b) Delete an element.
- c) Display the Queue.

5. Write a program in python to implement following operations on a Circular Queue.

- a) Insert an element.
- b) Delete an element.
- c) Display the Circular Queue.

6. Write a program in python to implement following operations on a Deque.

- a) Insert an element at front.
- b) Delete an element from front.
- c) Insert an element at rear.
- d) Delete an element from rear.
- e) Display the Queue.

7. Write a program in python to implement following operations on a Stack using linked List.

- a) Push an element.
- b) Pop an element.
- c) Display the stack.

8. Write a program in python to implement following operations on a Queue using Linked List.

- a) Insert an element.
- b) Delete an element.
- c) Display the Queue.

9. Write a program in python to implement following operations on a Binary Search Tree.

- a) Insert a node into the tree.
- b) Display the tree in In-order fashion.
- c) Display the tree in Pre-order fashion.
- d) Display the tree in Post-order fashion.

- e) Find out maximum element from the Binary Search Tree.
- f) Find out minimum element from the Binary Search Tree.
- g) Search whether a given element present in the Binary Search tree or not